

Quantum Leap: *China's Satellite and the New Arms Race*

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On August 2, 1939, at the dawn of World War II and six months after the discovery of uranium fission, Albert Einstein and Leo Szilard wrote a letter to U.S. President Franklin Delano Roosevelt. Building on Szilard's work, which had resulted in the first nuclear chain reaction and a patent on the first nuclear reactor with Enrico Fermi in the 1930s, Einstein warned Roosevelt that uranium could likely be used to build an atomic bomb. Even more worrying, as U.S. scientists wasted time doubting the potential of nuclear energy and weapons, Germany's program was advancing rapidly and the country already had access to Czechoslovakian uranium. It is this critical historical moment that the political theorist James Der Derian uses to frame Project Q, an ambitious research project, symposium series, and documentary that explores the international relations implications of a new generation of quantum technologies.

Nearly a century ago, the giants of physics were surely conflicted. Einstein was a noted pacifist. All were developing a field of research that fundamentally challenged Newtonian physics and the positivist certainty and existential grounding that it provided. The world, they discovered, was not logical, predictable, and measurable. At the smallest level, matter behaved differently. Atoms, photons, and electrons could be waves and particles at the same time. They could be connected to one another over vast distances. Although many saw disquieting randomness in this behavior, Einstein and the other leading physicists of the time debated the exciting possibilities for various configurations of this quantum matter. Their theories led to the development of nuclear fission and the astounding power that it promised. They also ultimately led to some of the great technological breakthroughs of the twentieth century, including transistors and lasers.

At the same time, the community of scientists saw the potential of these new discoveries as tools of war. Having lived through a war that saw new, experimental weapons unleashed to horrifying consequence, they were torn by the difficulties of reconciling their science with the potential impact of its application. Instead of calling for a ban on the use of this technology, however, engaged scientists lobbied the United States to weaponize it themselves as a counter to Nazi German advances. Acutely aware of the potential of what they were developing, and strategically prescient enough to share their knowledge with the United States, they urged Roosevelt to acquire uranium ore and establish a government nuclear weapons research program.

Initially, Roosevelt was cautious, establishing a “Uranium Committee” that purchased \$6,000 in graphite and uranium for the experimentation. A large-scale atomic project did not begin for two more years, until December 6, 1941, the day before Pearl Harbor. Within a year, the program had expanded to become the Manhattan Project, and as the scientists had anticipated, the nature of power in the twenty-first century would be largely defined by the development and influence of nuclear weapons.

And so is the case again as a new generation of quantum research is being developed in research labs around the world. Again, a scientific community is making striking breakthroughs in both theory and application, and once again, the implications to global peace and security are profound.

And as they did nearly a century ago, the great powers are taking note. The United States, China, Google, IBM, and others are all chasing the promise of a new generation of quantum science, steadily inching closer and closer to a true breakthrough. To give just one example, on August 16, the Chinese government launched the first quantum satellite, which will provide secure encrypted communication by teleporting information up to 1,200 kilometers (almost 750 miles). It works through a process known as “quantum entanglement,” whereby two particles—in this case, one on the ground and one in space—are held in a single superposition, allowing their properties to be connected. This allows for instantaneous communication that if measured, or observed, simply collapses, making it theoretically unhackable.

A CENTRALIZATION OF POWER

Most would agree that the digital age has, on balance, empowered the citizen. Through social media, unprecedented access to information, and collective action online, power has begun to shift away from traditional hierarchical organizations to decentralized networks of digital actors. However, several quantum technologies have the potential to reverse this advantage.

The first of these technologies, and the one shrouded in the most sci-fi fanfare, is quantum computing. In 1982, the famous physicist Richard Feynman proposed leveraging the “spooky” properties of quantum mechanics—mind-bending phenomena such as entanglement, wave-particle duality, and superposition—to build a new and far more powerful form of computer. In what quantum physicists lovingly term a “classical” computer, a bit of information exists in one of two distinct states: as a one or a zero. On the other hand, a quantum computer utilizes quantum bits, or qubits, which employ superposition to inhabit multiple states and allow for completely different types of calculations.

Working in qubits has the potential to vastly increase computational capacity. Where a conventional computer has to essentially go through possible outcomes one after another, a quantum computer could theoretically sift through massive amounts of information in search of shared properties, finding patterns in large data sets, for example.

The form of quantum technology that is probably the closest to operational is quantum communications, which has attracted a great deal of attention from states looking to keep their high-level correspondence safe from interception and espionage.

It is commonly noted that a quantum computer could theoretically run special algorithms that might be able to break public-key cryptosystems (such as the widely used RSA protocol), which rely on the premise that once extremely large prime numbers have been factored together, it becomes computationally unfeasible to reverse engineer the key. For example, it is not currently viable to brute force a 128-bit key by running all possible combinations, but MIT Professor Peter Shor's factorization algorithm—if successfully run on a quantum computer—would be able to break the key in polynomial time (read: fast). The development of quantum computing could therefore render the majority of online encryption vulnerable to those that held the technology. Although experts are divided about the risks of developing quantum, a National Security Agency (NSA) policy statement published in late 2015 caused quite a stir when it advised the private sector to move toward quantum-resistant encryption, seemingly suggesting that quantum computers may be more imminent than previously assumed. Indeed, there is a growing sense of urgency in the cryptographic community, because, as the University of Waterloo's Michele Mosca and colleagues have noted, the transition toward postquantum security will take a long time and a lot of preparation.

For years, observers have insisted that quantum computers are ages away, with optimistic estimates putting their earliest development at least 15 to 20 years off; and yet, as data collection efforts by nation-states and large corporations grow in ambition and scale, quantum computing provides a possible answer to a fundamental challenge. A quantum computer armed with Grover's search algorithm could be really good at finding shared properties—eventually perhaps being able to search through millions of e-mails or surveillance feeds. Imagine this capability paired with other emerging technologies, such as facial recognition, and the implications for state power in the international system are clear: these advances, and the actors who are likely to acquire them, could reverse the trends of digital empowerment that have been embedded in the history of the Internet.

Another particularly promising area is quantum location, which makes use of complex quantum-assisted sensing to provide resilient navigational and global-positioning capabilities. For example, the British Ministry of Defence is experimenting with a so-called quantum compass that would allow nuclear submarines to navigate without having to maintain satellite contact, potentially overcoming one of the primary challenges of stealthy deep-water navigation.

But the form of quantum technology that is probably the closest to operational is quantum communications, which has attracted a great deal of attention from states looking to keep their high-level correspondence safe from interception and espionage. In line with its satellite, the Chinese government is developing large-scale, theoretically unhackable quantum encryption projects, including the world's longest quantum communications network. These systems work by entangling photons, placing them in specific quantum states ("spins"), and using them to form a quantum key, which, owing to quirks of quantum mechanics, would change its behavior if intercepted by an eavesdropper (this also means that this form of communication would be quantum computer resistant, as it does not rely on conventional algorithms). If these projects continue to be successful, there are already plans to roll out an Asia-Europe quantum communications network in the next five years, providing a perfect foil to the potential development of quantum computing. It is unsurprising that U.S. rivals are aggressively pursuing quantum communications projects, which are almost certainly motivated by a desire to thwart pre- and postquantum digital espionage by the U.S. government.

But the prohibitively expensive nature of quantum communications networks could also create a two-tiered system, where only government correspondence is safe while that of ordinary citizens remains vulnerable to

interception by foreign governments (and their own). This highlights the real potential that second-generation quantum technologies will have a recentralizing effect, enabling states to reclaim some of the ground that they have lost to decentralizing digital technologies such as the World Wide Web and mobile networks.

A NEW OFFSET

During the administration of U.S. President Dwight D. Eisenhower, the U.S. government instituted its first “offset strategy”—a conscious attempt to use technology (in this case, nuclear weaponry) to overcome what it perceived as a strategic disadvantage in conventional military terms. After the Soviets developed their own nuclear capabilities and this advantage was negated, the United States moved toward their second offset, which used advances in GPS, surveillance, and communications as force multipliers.

In a speech delivered in December 2015, Deputy Secretary of Defense Robert Work outlined how the U.S. military’s historical military-technology edge was eroding once again and how it would have to collaborate with the private sector in its efforts to implement a game-changing “third offset.”

Quantum technologies could play a role in two ways: through new public-private partnerships and through dark money.

First, although DARPA (the agency of the U.S. Department of Defense that is responsible for the development of new technologies) has essentially classified quantum computing, communications, and assisted sensing projects, the private sector has pushed ahead through partnerships with government agencies. A joint NASA-Google quantum computing project, as well as many other initiatives, show that government agencies are collaborating. Quantum-related research is likely to remain hugely expensive, and Silicon Valley has an incentive to continue collaborating with Washington. It is possible that the real breakthroughs in quantum computing may be first developed in the private sector, by a company like IBM or Google, and government agencies are keen to ensure that these computers, if developed, do not fall into the “wrong” hands.

Meanwhile, a dark money arms race is likely already under way in the development of quantum technologies. Compare the potential of quantum technology with what we know of the current funding levels. The ability to break public-key encryption is a lever of tremendous commercial and state power. And yet, on the surface, the investment in quantum research is modest. To what extent are militaries and governments engaging in research we don’t know about?

As of right now, the investment in quantum technologies that we know about seems to be, conservatively, in the high hundreds of millions to low billions. For example, the Institute for Quantum Computing at the University of Waterloo, which is only one of many projects in this space, has received at least \$200 million in funding (\$50 million from the province of Ontario; \$50 million from the government of Canada; and \$100 million from a special fund investing exclusively in quantum technology, led by Mike Lazaridis, the founder of Research in Motion). It is probable that the major quantum research initiatives in the United States, such as NASA’s partnership with Google and Lockheed Martin’s partnership with the University of Southern California, have higher levels of investment (although the exact figures are not known). And in the same way that it was revealed in the Snowden leaks that DARPA was contemplating building its own quantum computer, it is almost certain that other states, such as China, have been pouring money into secret projects of their own.

At a recent symposium, the Q Symposium, the University of New South Wales’ Andrew Dzurak noted that “quantum research and development may have moved too far into the realm of commercial intellectual property and national security for global cooperation to be possible.” This is partially because the race for these

technologies has become part of a larger battle for international cyberdominance, and the development of fault-tolerant quantum computing (and its irresistible encryption-breaking potential) is too big a prize. Because of the tremendous cost and production timeline of quantum technologies, their development will be done by large research labs and funded by nation-states and large corporations. And so it is absolutely critical that Project Q explore the nature and limits of these alliances, what is being done outside the bounds of them, and what is being done in secret.

It is time to begin thinking through how the world will govern emerging quantum technologies.

OPENING THE BLACK BOX

The promise of quantum science has always been epistemological. It changes how and what we know. As a second generation of quantum technology comes online, three critical questions raised by and explored through Project Q are critical.

The first is whether quantum technologies will prove emancipatory or will reconcentrate power in the hands of states. At the Q Symposium, Professor Michael Biercuk, an experimental physicist and director of the Quantum Control Laboratory at the University of Sydney, pointed out that “new technology drives radical social change.” If we are going to take seriously the proposition that quantum could be disruptive, let alone emancipatory, then we need to ask who are the nimble outsiders developing these technologies to take on legacy institutions, and at what point will access to these technologies be democratized and available to the many in ways that challenge existing structures. It is far more likely that the early stages of the deployment of the technology will benefit incumbent actors.

Take the case of quantum positioning and quantum communications. On the one hand, these technologies have the potential to dramatically increase military capabilities. On the other hand, they could also profoundly empower individuals, providing new levels of privacy and agency if they trickle down into the public sector. For example, the tech journalist Patrick Tucker has suggested that quantum location technologies could potentially provide a replacement for the GPS in phones and hand-held devices, allowing them to run offline and perhaps keep the location data out of the hands of carriers or snooping government agencies. But power is often zero-sum. And it is worth assuming that the interests of those developing these technologies will determine who is empowered by them.

Observers also need to ask who is competing to get these technologies, and is there a tension between and within emerging strategic alliances. As Biercuk pointed out, the research has moved from “things to study to things to exploit,” meaning there will be real competition for capabilities that can be monopolized. There is a profound tension between the spirit of cooperation (the U.S. government and Silicon Valley, International Research labs) and the opportunities for strategic, scientific, and commercial gain: a confluence of interests that has led commentators to warn of an impending “quantum arms race.” We may have lost the window for a truly international project because the incentives for commercial and security gains are too strong. Along with the United States and China, Australia, Russia, and United Kingdom all are involved in the global race for quantum computing.

Third, and perhaps most important, it is time to begin thinking through how the world will govern emerging quantum technologies. In order to control the digital space, one needs both data and the tools to give them meaning. With meaning will come control and power, which opens up a wide range of governance challenges. According to Jairus Grove, director of the University of Hawaii’s Research Center for Futures Studies, quantum

technologies pose a “direct challenge to democratic decision-making and accountability.” As government agencies seek to collect “the whole haystack,” as the former NSA chief Keith Alexander once put it, and utilize increasingly algorithmically oriented forms of governance to rule their citizens, how do we ensure that even more opaque quantum algorithms are employed responsibly?

As a limited number of states and corporations seek fault-tolerant quantum technologies to exploit a decisive military advantage, they will surely change the ways in which we think about power and control in the international system. But even beyond shifts in power, so-called quantum social theory could be used to help researchers metaphorically and empirically understand social phenomena. In a new book on quantum theory, Alexander Wendt, a professor of political science at Ohio State, argues that although classical physics cannot explain concepts such as consciousness, perhaps thinking of collections of human minds as a quantum machine, and subject to the emerging scientific knowledge of quantum phenomena, can scientifically ground our understanding of social collectives. Quantum science could change how we know the world.

The first generation of quantum science unleashed not only the power of atomic weapons but new ways of understanding the universe. The scientists developing quantum technologies were actively engaged in heated debates about the moral responsibility of both. Project Q has sought to replicate this moment. As research continues at a breakneck pace, and as the hype around quantum technologies continues to escalate, it would be wise to not lose sight of the very tangible promise and peril that this new quantum era embodies—for much like the nuclear age, it may arrive sooner than we think.